# MLOPS Assignment 2

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## 1. Which run had the lowest latency, and why might that be?

Inthefirstexecution,**run_batch4**hadthelowestlatency(0.034s),followedbyrun_batch8(0.047s) and run_batch1 (0.163 s). Processing multiple sentences together reduces repeated inference-call overhead. The exact timings can vary between executions because of warm-up and system conditions.

## 2. Did accuracy change between runs? Why or why not?

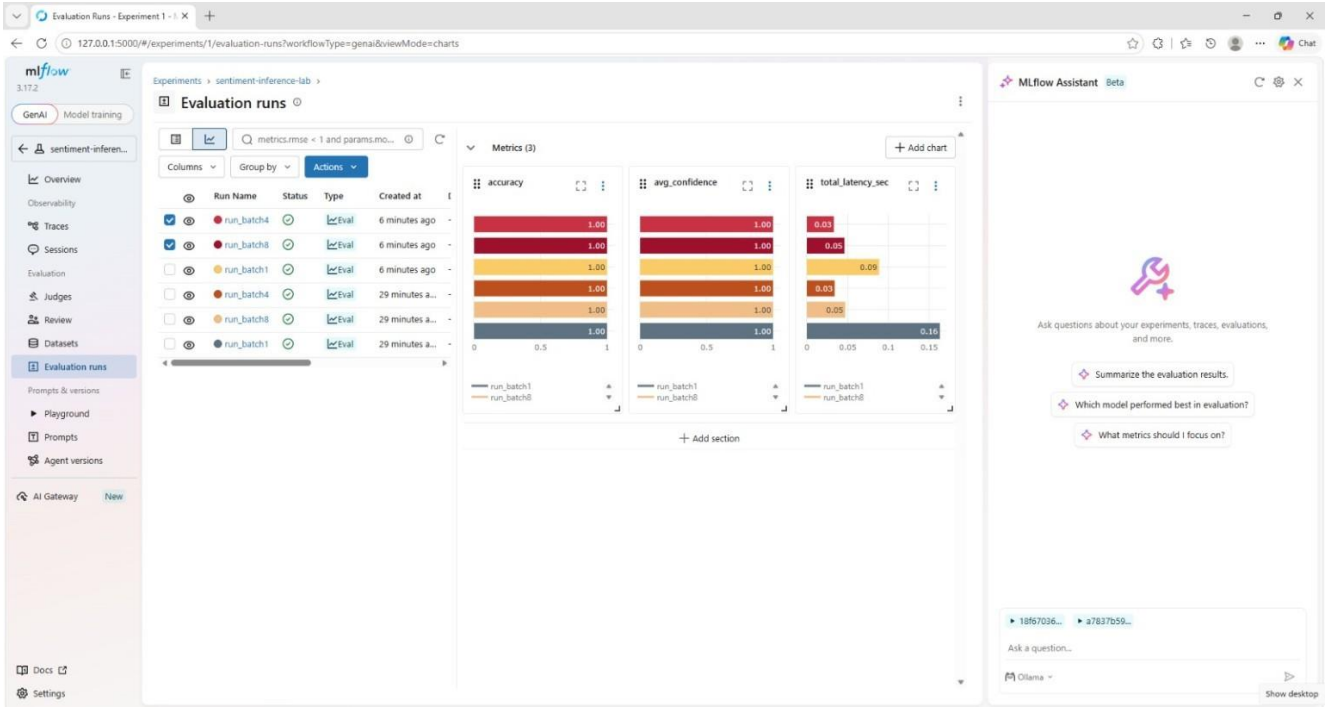
No.Accuracyremained**1.0(100%)**forallthreebatchsizes.Thesamepretrainedsentiment-analysis model and test sentences were used. Changing the batch size affects how inputs are grouped for inference, not the model weights or expected predictions.

## 3. Name two benefits of using MLflow here instead of printing results in the terminal.

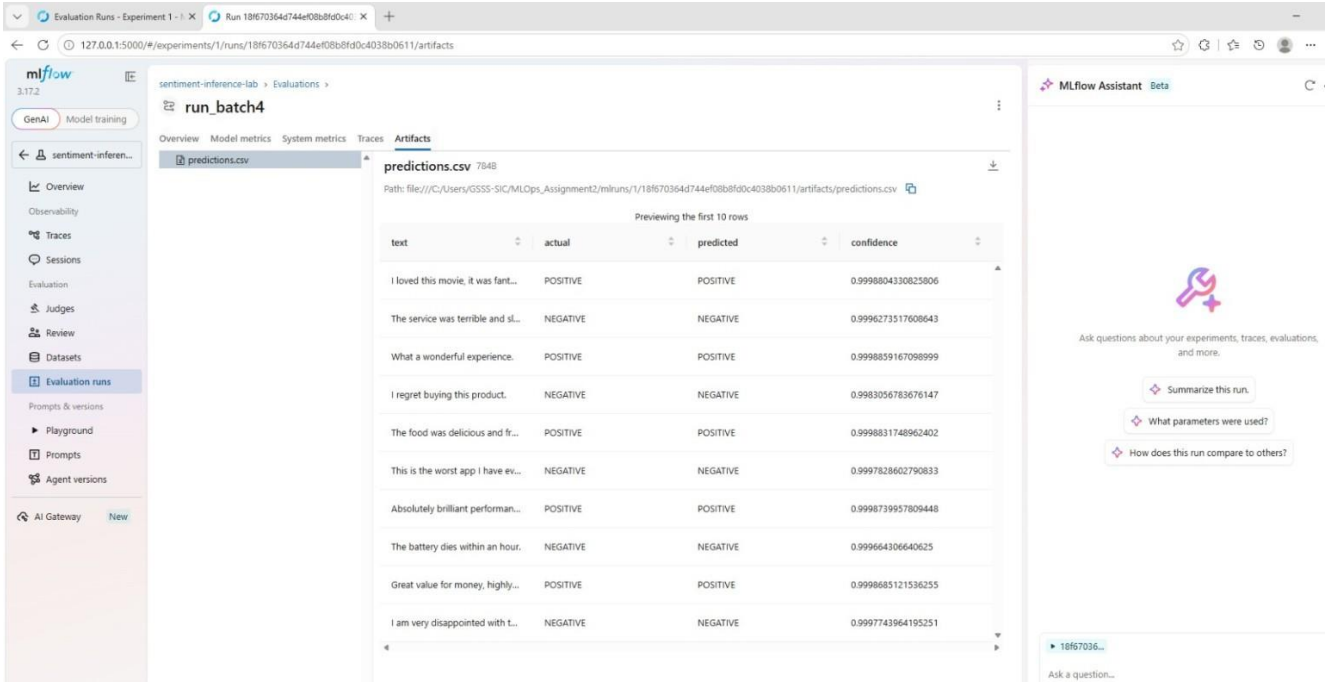
**Reproducibilityandrecord-keeping:**MLflowsavesparameters,metricsandartifactssuchas predictions.csv for later inspection.

**Easycomparison:**MLflowdisplaysrunsandtheirmetricssidebyside,makingiteasiertocompare accuracy, confidence and latency than reading terminal output alone.

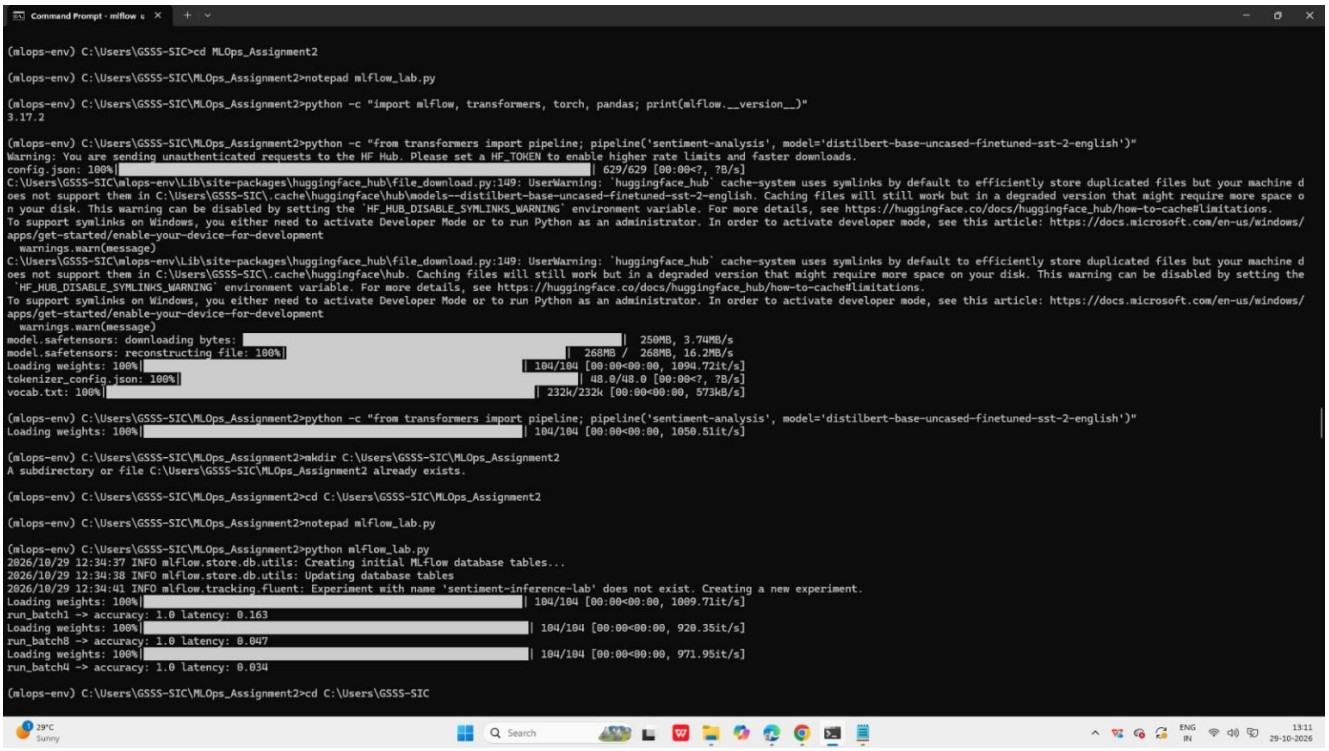
## Screenshot of the MLflow runs table and comparison charts:



## Screenshot of the run_batch4 artifact (predictions.csv):



## Execution: First run and recorded metrics:



## Execution: MLflow server and subsequent run:

